WC 2026 TOURNAMENT SIMULATOR

Complete Technical Report

An end-to-end machine learning pipeline predicting the FIFA World Cup 2026 using 5,000 Monte Carlo simulations

Historical Matches	Tournament Simulations	ML Models Trained	Teams Simulated
49,287	**5,000**	**4**	**48**
1916–2024	Monte Carlo	RF + Poisson	12 Groups

Predicted Champion: **Spain (12–15%)**

2026 | Data Science & Machine Learning

1. Project Overview

The WC 2026 Tournament Simulator is a full-stack data science project that predicts the outcomes of the 2026 FIFA World Cup. It combines historical football data spanning over a century (1916–2024), advanced feature engineering, four machine learning models, and a 5,000-iteration Monte Carlo simulation engine — all deployed as an interactive web application accessible to any user worldwide.

1.1 Project Goals

- → Predict group stage advancement probabilities for all 48 qualified teams across 12 groups
- → Generate knockout bracket predictions from Round of 32 through the Final
- → Compute overall championship probabilities for every team with 95% confidence intervals
- → Build an interactive Streamlit app for exploring predictions, alternate scenarios, and personal bracket picks
- → Deploy publicly on Hugging Face Spaces for instant access with no cold-start delays

1.2 Technical Stack

Layer	Technology	Purpose
Data Processing	Python, Pandas, NumPy	ETL, cleaning, feature engineering
Machine Learning	Scikit-learn, Random Forest, Poisson	Match prediction, goals, penalties
Simulation Engine	Monte Carlo, NumPy Poisson sampling	5,000 full tournament iterations
Web Application	Streamlit	7-page interactive dashboard
Deployment	StreamlitCloud	Always-on public link, zero cold starts
Version Control	GitHub	Source code & CSV data hosting

1.3 Application Architecture

The project follows a clean three-layer architecture:

- **Layer 1** Colab Notebook: Raw data → ETL → Feature engineering → Model training → Monte Carlo → CSV exports
- **Layer 2** Services Layer (Python): bracket_service.py, prediction_service.py, results_service.py — load CSVs, provide query methods
- **Layer 3** Frontend Layer (Streamlit): frontendapp.py, bracket_page.py, scenarios_page.py — 7 interactive pages

2. Data Processing & ETL Pipeline

2.1 Raw Datasets

Dataset	Records	Description
results.csv	49,287 matches	International match results 1916–2024
shootouts.csv	675 entries	Penalty shootout records with winner/loser
goalscorers.csv	47,601 records	Individual goal scorer data per match
former_names.csv	36 entries	Team name standardization mappings
fifa_ranking.csv	67,472 records	Monthly FIFA rankings 1993–2024

2.2 Data Cleaning & Standardization

- → Team name standardization using former_names.csv — e.g. 'Cabo Verde' → 'Cape Verde', 'Czechia' → 'Czech Republic'
- → All date columns converted to datetime format for temporal feature extraction
- → Shootout data merged into match records to flag penalty decisions
- → Goal scorer data aggregated per match: unique scorers, penalty goals, own goals

2.3 Late Tournament Filter

Raw data was filtered to retain only high-stakes competitive matches from major tournaments, yielding 3,559 matches:

Tournament	Competition Level	K-Factor Used
FIFA World Cup	Highest	60
UEFA Euro	Very High	48
Copa América	Very High	38
African Cup of Nations	High	28
AFC Asian Cup	High	24
CONCACAF Gold Cup	High	18
OFC Nations Cup	High	18

2.4 Key EDA Insights

Metric	Value	Implication
Average goals per match	2.67	Low-scoring — defensive play dominates
Clean sheet rate	52.8%	Over half of games feature a clean sheet
Home win % (neutral venue)	42.4%	Home advantage is substantially weaker
Home win % (home venue)	46.3%	Confirms neutral flag as key feature
Average unique scorers	~2 per match	Attacking depth matters
Shootout conversion rate	69.1%	Zone 3 (bottom right) most reliable
Golden Boot avg (modern era)	6.3 goals	Realistic tournament scoring target
Elo gap: #1 to #48	328 points	Spain (1,777) vs Canada (1,449)

3. Feature Engineering

Over 30 predictive features were engineered across four categories to capture team quality, recent performance, tournament experience, and match context.

3.1 Feature Categories

Category	Features Created	Key Detail
Elo Ratings	home_elo, away_elo, elo_diff	Time-weighted, tournament-adjusted K-factors
FIFA Rankings	home_rank, away_rank, rank_diff, points_diff	Blended 60/40 with 2026 anchor ranking
Recent Form (5 & 10 matches)	win_rate, gs_avg, gc_avg, clean_sheet_rate	Missing values filled by rank anchors
WC Pedigree	home_wc_pedigree, away_wc_pedigree, pedigree_diff	Titles×3 + Finals×1.5 + Semis×0.5
Composite Strength	strength_diff	55% Elo + 45% FIFA Rank, normalized
Match Context	neutral, is_penalty_shootout, match_importance	Context flags for situational adjustment

3.2 Elo Rating System

Elo ratings were calculated exclusively on late-tournament matches using a time-weighted system:

- → Full weight for matches in the last 4 years
- → 75% weight for matches in years 5–8
- → 50% weight for older historical matches
- → Tournament-specific K-factors: World Cup (60) > Euros (48) > Copa América (38) > Regional (18–24)

3.3 Most Important Feature: Elo Difference

Elo difference alone accounted for 16.6% of the Random Forest's feature importance — higher than any other single variable. The rank difference contributed 10.8%, confirming that relative team quality is the dominant predictor of match outcomes in high-stakes football.

4. Machine Learning Models

4.1 Model Overview

Model	Type	Target	Performance
Match Outcome	Random Forest (3-class)	Home Win / Draw / Away Win	Accuracy: 60.3% | Log Loss: 0.8486
Home Goals	Poisson Regression	Expected home goals	MAE: 0.96 goals
Away Goals	Poisson Regression	Expected away goals	MAE: 0.74 goals
Penalty Kicks	Random Forest (binary)	Goal scored / Saved	Accuracy: 71.9% (individual kicks)

4.2 Match Outcome Model (Random Forest)

- → Trained on 1,336 pre-2022 matches with 16 features
- → Validated on the full 2022 World Cup: 116 matches
- → Expected Calibration Error (ECE): 0.09 — well-calibrated probabilities
- → Conservative predictions in 45–60% range — ideal for Monte Carlo sampling
- → Selected over Logistic Regression and XGBoost for best calibration on tournament data

4.3 Goals Models (Poisson Regression)

- → Separate models for home and away expected goals (xG)
- → Poisson distribution validated: variance/mean ratio = 1.18 (appropriate for goal scoring)
- → Home goals MAE: 0.96 | Away goals MAE: 0.74 — 3–11% better than baseline averages
- → Features include Elo ratings, rank differentials, recent attacking/defensive form

4.4 Penalty Shootout Model (Random Forest)

- → Trained on 256 individual penalty kicks from historical data
- → 71.9% accuracy at predicting individual kick outcome
- → Shootout winner accuracy: 57.1% vs 50% baseline — meaningful predictive power
- → Top features: penalty number (16.2%), keeper mismatch (13.4%), shot zone
- → Team-specific shooting tendencies tracked for 26 teams with sufficient history

4.5 Key Finding: Elo Alone Beats Elo + Form

On the 2022 World Cup validation set, the Elo-only model outperformed models incorporating recent form features. This suggests that recent form adds noise rather than signal in short tournament contexts where form volatility is high and sample sizes are small. The final model includes form features but relies primarily on Elo and ranking differentials.

5. Monte Carlo Tournament Simulation

5.1 Simulation Parameters

Parameter	Value
Total simulations	5,000 complete tournaments
Teams	48 (official WC 2026 draw)
Groups	12 groups of 4 teams
Group stage matches	72 per simulation
Knockout matches	33 (R32 + R16 + QF + SF + 3rd place + Final)
Total match simulations	~520,000
Precomputed matchups (cache)	2,256 (all 48×47 ordered pairs)
Simulation speed	~2,000 simulations/second

5.2 Cache-Based Optimization

The core optimization enabling 2,000 sims/second is the MATCH_CACHE — a precomputed dictionary of all 2,256 possible team pairings containing:

- → Home win probability, draw probability, away win probability
- → Expected home goals (lambda_home), expected away goals (lambda_away)
- → Computed once before simulation; referenced instantly during each run

This eliminates repeated ML model calls during simulation. Without the cache, each match would require a full Random Forest inference, reducing speed by 10–20×.

5.3 Simulation Flow

Stage	Method	Tiebreaker
Group Stage	72 Poisson-sampled matches per group	Points → Head-to-head → Goal diff → Goals scored → Fair play
Best 8 Thirds	Ranked by points, goal difference, goals scored	Cross-group comparison
R32 → R16 → QF → SF	Probabilistic sampling from MATCH_CACHE	Extra time (70% xG) → Penalty model
3rd Place Match	SF losers face each other	Penalty model if tied
Final	SF winners face each other	Penalty model if tied

5.4 Critical Bug Fixes Applied

- **Fix 1** Match 79 bracket correction: runner-up of Group A (not winner) — prevented ghost winners appearing in wrong matches
- **Fix 2** Winner validation at every round: winner must be in {team1, team2} — eliminates impossible results
- **Fix 3** Same-team detection: prevents a team playing itself due to bracket corruption
- **Fix 4** Advancement probability cap at 100%: eliminates impossible values from normalization errors
- **Fix 5** Dynamic cascade simulation: later rounds derive teams from actual winners, not hardcoded CSV values

6. Tournament Results

6.1 Championship Probabilities (Top 15)

Rank	Team	Champion %	Finalist %	Semi-Final %	Quarter-Final %
1	Spain	12–15%	22–26%	38–42%	58–63%
2	Argentina	11–13%	21–25%	36–40%	56–61%
3	France	10–12%	19–23%	34–38%	54–59%
4	Brazil	8–10%	16–20%	30–34%	50–55%
5	England	7–9%	14–18%	27–31%	46–51%
6	Germany	6–8%	12–16%	24–28%	43–48%
7	Netherlands	5–7%	11–14%	21–25%	40–45%
8	Portugal	4–6%	9–12%	18–22%	36–41%
9	Belgium	3–5%	7–10%	15–19%	32–37%
10	Colombia	2–4%	5–8%	12–16%	26–31%
11	Uruguay	2–3%	4–6%	9–13%	20–25%
12	Morocco	1–3%	3–5%	8–12%	18–23%
13	Senegal	1–2%	2–4%	6–10%	15–20%
14	Japan	1–2%	2–3%	5–9%	13–18%
15	South Korea	<1%	1–2%	4–7%	11–15%

6.2 Most Likely Final

🇪🇸 **SPAIN**
12–15% champion probability

🇦🇷 **ARGENTINA**
11–13% champion probability

6.3 Group Analysis Highlights

Group	Teams	Most Likely to Advance	Competitiveness Score
Group H	Spain, Cape Verde, Saudi Arabia, Uruguay	Spain + Uruguay	8.2 (most predictable)
Group A	Mexico, South Africa, South Korea, Czech Republic	South Korea + Mexico	14.1

Group	Teams	Most Likely to Advance	Competitiveness Score
Group C	Brazil, Morocco, Haiti, Scotland	Brazil + Morocco	11.8 (most competitive)
Group D	USA, Paraguay, Australia, Turkey	USA + Turkey	12.3 (most competitive)

6.4 Upset Probability Highlights

The highest upset probabilities occur when mid-tier European or African teams face top South American sides in Round of 32 fixtures:

- → Switzerland vs Netherlands: ~38% upset (Swiss overperform vs top-10 teams historically)
- → Morocco vs any top European: ~35% upset probability in R16/QF (proven WC 2022 pedigree)
- → Colombia vs any SF opponent: ~32% upset — the 'dark horse' of the tournament

7. Streamlit Web Application

7.1 Application Pages

Page	Description	Key Feature
Champion Probabilities	Top 15 most likely champions with 95% CI	Color-coded bar chart + medal-style top 3
Group Stage	Position probabilities for all 12 groups	Stacked bar + competitiveness analysis
Knockout Bracket	Most likely path to the Final	Text bracket tree + match-by-match breakdown
Matchup Predictor	Head-to-head for any 2 teams	Win %, draw %, xG from precomputed cache
Alternate Realities	6 scenario simulations with strength slider	Re-roll any match, save favorite brackets
Interactive Bracket	Full user bracket with cascading picks	Auto-updates downstream, champion banner
Team Stats	Elo, FIFA rank, stage survival curves	Per-team probability progression chart

7.2 Alternate Realities — 6 Scenarios

Scenario	Modifier	Effect
Most Likely (Default)	0 (deterministic)	Reads CSV directly — same result every time for every user
Giant Killers (Upsets)	+18% underdog boost	Underdogs drastically overperform — Cinderella stories
European Dominance	−12% underdog boost	Favorites win more often — European teams ascendant
South American Revival	+12% underdog boost	South American teams exceed their rankings
Home Advantage	+10% underdog boost	CONCACAF teams boosted by crowd support
Golden Generation	−8% underdog boost	Slight favorites edge — star players deliver

7.3 Performance Optimizations

- → Single data load: all CSVs loaded once in app.py with @st.cache_data — never re-read by sub-pages

- → DataFrames passed as function arguments — eliminates duplicate memory usage across pages
- → Most Likely scenario reads from CSV only — never re-simulates, identical for every user
- → @st.cache_data on R32 extraction and Most Likely bracket build — computed once per session
- → Dynamic cascade: later rounds computed from simulation winners, not hardcoded CSV rows

8. Deployment & Infrastructure

8.1 Deployment Platform: Hugging Face Spaces

The application is deployed on Hugging Face Spaces using Docker SDK — chosen specifically because:

- ✓ No cold starts: always running, opens in seconds for any user worldwide
- ✓ Free forever for public spaces — no cost compared to Render/Railway
- ✓ Docker SDK supports Streamlit natively on port 7860 (the mandatory HF port)
- ✓ Auto-deploys when GitHub repo is pushed — zero manual intervention

8.2 Project Structure

Directory/File	Purpose
app.py	Main Streamlit entry point — loads all data, routes pages
frontend/frontendapp.py	7-page dashboard logic
frontend/bracket_page.py	Interactive user bracket with cascade picks
frontend/scenarios_page.py	Alternate reality simulation engine
services/bracket_service.py	Bracket builder, cascade logic, matchup lookup
services/prediction_service.py	ML model loading and inference
services/results_service.py	CSV data loading and query methods
config/tournament_config.py	Groups, FIFA ranks, bracket pairings, constants
data/*.csv	All simulation outputs (champion probs, matchups, knockout bracket, etc.)
models/*.pkl	Trained ML models, scalers, Elo ratings, match cache
Dockerfile	HF Spaces Docker config — runs Streamlit on port 7860
requirements.txt	Minimal dependencies: streamlit, pandas, numpy, plotly, scikit-learn

9. Critical Bugs Fixed During Development

Bug	Root Cause	Fix Applied	Impact
Spain winning Qatar vs Iran	match 79 used 'winner' instead of 'runner-up' of Group A in bracket_structure	Changed one word: 'winner' → 'runner_up' in both bracket definitions	Eliminated ghost winners across all simulations
Ghost winners in scenarios	simulate_full_tournament read all rounds from CSV — Spain always in M84/M93 regardless of who actually won	Only R32 from CSV; R16/QF/SF/Final from simulation winners dict	Bracket now self-consistent per simulation
Matchup Predictor 'not found'	matchup_probabilities.csv only had top 20 teams (380 rows)	Rebuilt CSV with all 48×47=2,256 team pairs; added reverse lookup	All team combinations now work
Blank pages on Streamlit Cloud	Same CSVs loaded 3–4× by different modules — exceeded 1GB RAM on HF	Single load in app.py, data passed as function arguments	All 7 pages load cleanly
Third place wrong teams	match 104 read from CSV slot instead of actual SF losers	Derived from SF results[101]['loser'] and results[102]['loser']	Third place always between real semi-final losers
Wrong knockout_bracket.csv	Probability = per-tournament freq not per-match win rate (France 10.7% was 'won the tournament', not 'beat opponent')	Rebuit: prob = wins among sims where THIS pair played	Probabilities now meaningful (45–75% range)

10. Conclusions & Future Work

10.1 Key Findings

- → Spain is the most likely champion (12–15%) but no team exceeds 15% — reflecting genuine tournament uncertainty
- → Elo difference is the single most predictive feature (16.6% importance), outperforming recent form
- → Poisson distribution is appropriate for football goal modeling (variance/mean ratio = 1.18)
- → 5,000 simulations provide stable probability estimates with ±2–3% confidence intervals for top contenders
- → The 2022 World Cup validation confirms Random Forest as the best-calibrated model for this task

10.2 Model Limitations

- → Player availability and injuries are not modeled — a key source of real-world uncertainty
- → Form features add noise on WC validation set — short tournament context reduces their signal
- → Penalty model accuracy (57.1% at shootout level) reflects the high randomness of penalties
- → Historical data underrepresents some AFC/CAF teams due to lower international match frequency

10.3 Future Improvements

- → Player-level simulation for Golden Boot: distribute team goals by individual scoring probability
- → Real-time pre-tournament betting odds integration for model calibration validation
- → Enhanced tiebreaker: historical card rates per confederation instead of Elo-based lot drawing
- → User-editable team strengths: allow manual Elo adjustments in the Alternate Realities page
- → Golden Glove simulation using clean sheet rates and penalty save probabilities

Work done by Tennich Brahim : simulation

Mail, Linkedin , Github

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